Part 3 — Cross-Language Comparison

Selected patterns: Repository, Strategy and Observer. Each was implemented in Java, Python, JavaScript and C++ using the same conceptual problem and expected output within each pattern family.

Pattern	Java ms	Python ms	JavaScript ms	C++ ms
Repository	24.629	594.444	21.222	1.105
Strategy	26.806	567.115	21.456	1.117
Observer	56.745	571.284	22.141	1.356

All 12 implementations passed. Timing protocol: compile first, one warmup, five measured process runs, report median. Timing includes process startup and test execution.

Source LOC ranged from 3–15 and maximum function cyclomatic complexity from 1–2. Java uses explicit interfaces/classes; Python and JavaScript use lighter-weight class/object structures; C++ uses explicit compiled classes.

These are process-level measurements of small deterministic snippets, not production-language benchmarks.